

Frontend Theory Assignment 2

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Subject: Frontend Development

1. What is CSS Box Model?

The CSS Box Model is basically a rectangular box that wraps around every single HTML element on a webpage. It dictates how elements are structured, spaced, and laid out on the screen. It is made up of four main layers from the inside out:

- **Content:** The actual text, images, or media inside the element.
- **Padding:** The transparent inner space directly surrounding the content, keeping it away from the border.
- **Border:** A visible outline wrapping around the padding and content.
- **Margin:** The outermost transparent space that pushes other surrounding elements away.

2. What is media query? Explain with a sample code.

A media query is a fundamental CSS3 feature used to build responsive websites. It allows us to apply specific blocks of CSS styles only when certain conditions are met, such as when the screen width drops below a particular size (e.g., switching from desktop view to mobile view).

```
/* Default style for desktop or large screens */
body {
  background-color: #e0f2fe;
}

/* Style applied when the screen width is 600px or smaller */
@media screen and (max-width: 600px) {
  body {
    background-color: #fce7f3;
  }
}
```

3. What are the features of Bootstrap?

Bootstrap is one of the most popular open-source front-end frameworks used for designing modern, responsive websites quickly. Its key features include:

- **Mobile-First Design:** Styles are optimized for mobile phones first and scale gracefully up to larger desktop screens.
- **Flexible Grid System:** A robust 12-column system making responsive layout creation very straightforward.
- **Pre-styled Components:** Ready-to-use user interface pieces like buttons, navigation bars, modals, cards, and dropdowns.
- **JavaScript Plugins:** Built-in components for interactive features like carousels, tooltips, and alerts without heavy setup.

4. Write an example of a basic grid structure in Bootstrap?

Bootstrap's grid system uses a series of containers, rows, and columns to layout and align content. It is built with flexbox and is fully responsive. Below is an example of a simple 3-column layout:

```
<div class="container">
  <div class="row">
    <div class="col-md-4 bg-light p-3">Column 1 (Width 4)</div>
    <div class="col-md-4 bg-secondary text-white p-3">Column 2 (Width 4)</div>
    <div class="col-md-4 bg-light p-3">Column 3 (Width 4)</div>
  </div>
</div>
```

5. What are CSS Selectors?

CSS selectors are patterns used to locate and "select" specific HTML elements on a webpage so that we can style them. The most common types include:

- **Element Selector:** Selects tags directly (e.g., `p { color: blue; }`).
- **Class Selector:** Selects elements assigned with a specific class name, denoted by a dot prefix `.` (e.g., `.box { padding: 10px; }`).
- **ID Selector:** Selects a unique element by its ID, denoted by a hash prefix `#` (e.g., `#header { background: black; }`).

6. What are CSS backgrounds, list the properties?

CSS background properties are used to add visual effects like colors, gradients, and images to the background of HTML elements. Here are the core background properties:

- `background-color` : Sets the solid background color of an element.
- `background-image` : Sets one or multiple background images.
- `background-repeat` : Controls if and how the background image repeats (e.g., `repeat-x` , `no-repeat`).
- `background-position` : Sets the starting location of the background image (e.g., `center` , `top left`).
- `background-size` : Specifies the dimensions of the background image (e.g., `cover` , `contain`).
- `background-attachment` : Determines whether the background scrolls with the page or remains fixed.

7. How can we overlap elements in CSS?

Overlapping elements in CSS can be achieved using a few different methods depending on the layout requirement:

- **Positioning and z-index:** By setting an element's position to `absolute` or `relative` , we can use the `z-index` property. Elements with a higher numeric `z-index` appear on top of those with a lower value.
- **CSS Grid:** Placing multiple items into the exact same grid cell or overlapping grid areas stacks them automatically.
- **Negative Margins:** Applying negative margins (e.g., `margin-top: -20px;`) shifts an element upward, causing it to overlap the preceding element.

8. What are the different positioning properties in CSS?

The CSS `position` property dictates how an element is laid out and positioned in the document flow. There are 5 main positioning values:

- **Static:** The default behavior. Elements flow naturally in the document order. Offset properties (top, left, etc.) do not apply.
- **Relative:** Positioned relative to its normal starting position. Offsets move it without disrupting the space around it.
- **Absolute:** Positioned relative to its closest positioned ancestor. It is completely removed from the normal document flow.
- **Fixed:** Positioned relative to the browser window (viewport). It stays glued in place even when the user scrolls the page.
- **Sticky:** Acts like relative until the page hits a specific scroll threshold, after which it becomes fixed in place.

9. What is ECMAScript?

ECMAScript (often abbreviated as ES) is the official scripting language specification published by Ecma International. It acts as the underlying standard and blueprint that JavaScript is built upon. While JavaScript is the practical programming language developers write, ECMAScript defines the syntax, rules, and features (such as arrow functions, classes, and async/await) that JavaScript engines must implement.

10. What is the difference between var, let and const in JavaScript? Explain Temporal Dead Zone?

Differences:

- **var:** Function-scoped, can be re-assigned, and allows redeclaration. It gets hoisted to the top and initialized with `undefined`.
- **let:** Block-scoped `{}`, can be re-assigned, but cannot be redeclared in the same scope. Hoisted but not initialized.
- **const:** Block-scoped `{}`, cannot be re-assigned or redeclared. Must be assigned a value immediately when declared.

Temporal Dead Zone (TDZ): The TDZ is the time period between when a block scope is entered and when a `let` or `const` variable is actually declared and initialized in code. During this period, the variable exists in memory (hoisted), but attempting to access it results in a `ReferenceError`. The zone ends as soon as execution reaches the line where the variable is defined.